

Roll No.

Total Pages : 3

CMCS/M-22

24547

CLOUD COMPUTING AND IOT

Paper–MS-20-44

Option–(iii)

Time Allowed : 3 Hours]

[Maximum Marks : 75

Note : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

Compulsory Question

1. Answer the following questions in brief : $5 \times 3 = 15$
- (a) What is cloud computing? Explain five characteristics of cloud computing.
 - (b) What are compute services on cloud computing? Explain in brief.
 - (c) What is IoT? Explain any two applications of IoT.
 - (d) What are privacy issues in IoT? Explain.
 - (e) What is storage-as-a-service? Explain.

UNIT–I

2. (a) Discuss the evolution of cloud computing to current stage in brief. 7

- (b) What are underlying technologies of cloud computing? Discuss each in brief. 8

3. (a) What do you mean by virtualization? Explain benefits and drawbacks of virtualization. 7

- (b) What are CPU and network virtualization? Explain. 8

UNIT–II

4. (a) What are database services? Explain any one database service in detail. 8

- (b) What are threats and security issues in cloud computing? Explain in brief. 7

5. Write short notes on the following :

- (a) Content delivery services.

- (b) Analytics services. 8

UNIT–III

6. (a) What should reference architecture cover in IoT general conceptual frame work? Explain. 8

- (b) Explain six layer modified OSI model for IoT/M2M with suitable diagrams. 7

7. (a) Explain different types of Bluetooth wireless technologies used in IoT communication. 7

- (b) What is RF transceiver? Discuss different modules of RF. 8

UNIT-IV

8. (a) What are lightweight M2M communication protocol?
Explain. 8
- (b) What are constrained communication protocol? Explain.
7
9. Write short note on the following message communication
protocols for connected devices :
- (a) CoAP-MQ 7
- (b) XMPP. 8